



S17 – River Flow Simulator

The Armfield S17 River Flow Simulator is a professionally built tank to demonstrate and simulate the formation of river features

PRODUCT CODE: S17

Categories: Educational, Environmental, Geology & Geography, S Series - Applied Hydraulics & Hydrology, S Series - Hydraulics & Hydrology, S17 - River Flow Simulator

Tags: Advanced Concepts , Bed Load Motion , Bedload Motion Capabilities , Braided Channels , Channel Morphology , Channelisation , Cutbanks , Deposition , Erosion , Flow , Fluvial Geomorphology , Gravel Mining , Hydraulics & Hydrology , Hydrographs , Meandering Channels , Meandering Water Courses , Model Behaviour , Outlet Weir , Research Purposes , Riffles , River Features , River Flow Simulator , S Series , S17 , Sand Point Bars , Sediment Transport , Slope , Stilling Mechanism , Straight Channels , Stream Power , Stream Response , Thalweg

DESCRIPTION

The Armfield S17 River Flow Simulator is a professionally built tank to demonstrate and simulate the formation of river features, including flow and bed load motion. It makes an excellent introduction to the study of fluvial geomorphology and can also demonstrate sophisticated and advanced concepts for research purposes.

The unit comprises a rectangular tank, made of stainless steel, measuring 215cm by 90cm. The tank is positioned and secured on a metal supporting frame. A manual jacking system enables the slope of the tank to be easily and safely adjusted, even when the tank is full of material and water.

The sturdy construction enables real sand to be used in the tank. This gives fine detailed and realistic feature simulation not obtainable on systems which use plastic pellets. In this way all principles of river formation and bed load motion can be studied. For best results the use of washed and graded sand is recommended.

The water reservoir is located at the tank discharge and water is pumped from the reservoir to the water inlet vessel using a low voltage submersible pump.

The inlet vessel is a 1.8l vessel, which can be positioned anywhere in the main tank. It is kept in its desired position by the sand within the tank.

The vessel incorporates a stilling mechanism and an outlet weir designed to produce a smooth output water flow into the main tank.

The system provides complete adjustment of water flow using a variable area flow meter and manual ball valve, which are mounted to the tank.

The water then flows through the sand where the formation of meandering channels, point bars, cut banks, riffles, thalwegs etc. can be observed and analysed.

At the end of the tank there is an outlet standpipe to return the water to the reservoir. The height of the standpipe is adjustable to enable different outlet conditions to be explored.

The system is also supplied with a sand scoop, scraper and surface profiling system, comprising a crossbeam and depth gauge.

The tank can easily be removed from the frame and the frame can be dismantled for storage using simple tools.

TECHNICAL SPECIFICATIONS

Working area: 2150mm x 900mm

Flow range: 1.5-15 l/min

Accuracy of flow metering: $\pm 2\%$ full-scale

Slope adjustment – working slope: 0-6°, 10° max

FEATURES & BENEFITS

- Robust stainless steel tank
- Uses sand for realistic and highly detailed feature formation
- Sturdy frame with dual jacks to vary slope
- Stilled water inlet vessel featuring variable positioning
- Variable height exit
- Self-contained, includes pump and water reservoir
- Variable flow with flow meter

Experimental content

- Experimental investigation on erosion and deposition
 - Characteristics of meandering water courses – the effects of changing stream power on channel morphology
 - Sediment transport and bedload motion capabilities
 - Determination of hydrographs from model behaviour
 - Channel morphology studies
 - Channel stability
 - Effect of base level change
 - Demonstrating stream response to channelisation
1. Formation of river features and development over time:
 - erosion
 - deposition
 - thalweg
 - cutbanks
 - sand point bars
 - riffles
 - straight, braided and meandering channels
- Gravel mining demonstration



Ordering specification

- A self-contained river flow simulator for the study and visualisation of river formations, bedload studies and fluvial geomorphology
- Stainless steel tank with a 2150mm x 900mm working area
- Tank mounted on a steel frame with jacking system to easily and safely adjust the tank slope while fully loaded
- Uses sand as the media for detailed feature development and bedload motion studies
- Water inlet vessel with stilling can be positioned anywhere in the tank
- Water reservoir, submersible pump, variable area flow meter and control valve giving flows from 1.5 to 15l/min
- Supplied with comprehensive instruction manual



Requirements

Electrical supply:

- 24V dc at 2.5A max
- A universal adaptor is supplied for operation from ac mains

Cold Water

Shipping Specifications

PACKED AND CRATED SHIPPING SPECIFICATIONS

- Volume: 1.4m³
- Gross Weight: 120kg

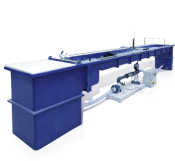
Dimensions

- Length: 2.60m
- Width: 1.06m
- Height: 1.02m

Order Codes

- S17-UK
- S17-EU
- S17-B

YOU MAY ALSO LIKE...

			
S8-MKII – SEDIMENT TRANSPORT DEMONSTRATI ON CHANNEL	S2 – MOBILE BED AND FLOW VISUALISATION TANKS	S1 – DRAINAGE AND SEEPAGE TANK	S12-MKII-50 – ADVANCED ENVIRONMENT AL HYDROLOGY SYSTEM
Read more	Read more	Read more	Read more

Armfield Limited. Unit 10 Headlands Business Park, Salisbury Road, Ringwood, Hampshire BH24 3PB.
England. Tel: +44(0) 1425 478781

